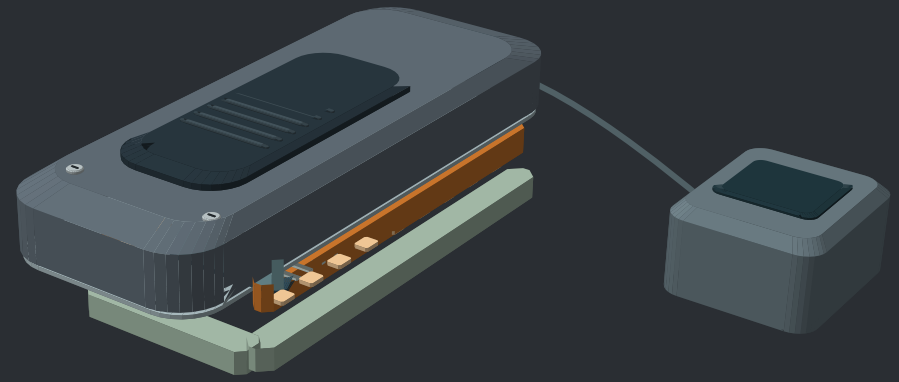
VASCOUT

Research discussion: CYBERDYNE / Acoustic X



ILLUSTRATIVE PROBE + SEPARATE ACQUISITION UNIT

A proposed ultrasound-led vascular imaging system for clinician-controlled assessment. Tailored research prospect briefing.

PROSPECT ONLY / INTEREST UNCONFIRMED / NOT SENT

Explore the live concept and evidence

Difficult access deserves clearer evidence.

Research intent: test whether anatomy and image-quality feedback can support clinician assessment.

CLINICAL CONTEXT / S10a + S10b

Difficult peripheral access

Published ultrasound-access findings differ by workflow and operator. The comparator for VASCOUT should be ultrasound alone, with a defined task and ground truth.

FIELD RESEARCH / UNVERIFIED

Low perfusion. Limited resources.

Dehydration, shock, motion, gloves and limited resources motivate proposed studies. VASCOUT benefit under these conditions has not been measured.

The intended contribution

Make anatomy and uncertainty visible for review. The concept does not select an insertion site, prescribe treatment or demonstrate successful IV access.

S10a / SOURCE S10b / SOURCE S07 / SOURCE

Ultrasound first. Added contrast to test.

Visible transillumination, NIR camera imaging and photoacoustic tomography are distinct methods. [S01]

01 / ACQUIRE

Ultrasound structure

Candidate linear array; coupling and low contact force.

>

02 / COMPARE

Experimental PA contrast

Test US alone against US + PA under matched conditions.

>

03 / REVIEW

Anatomy + uncertainty

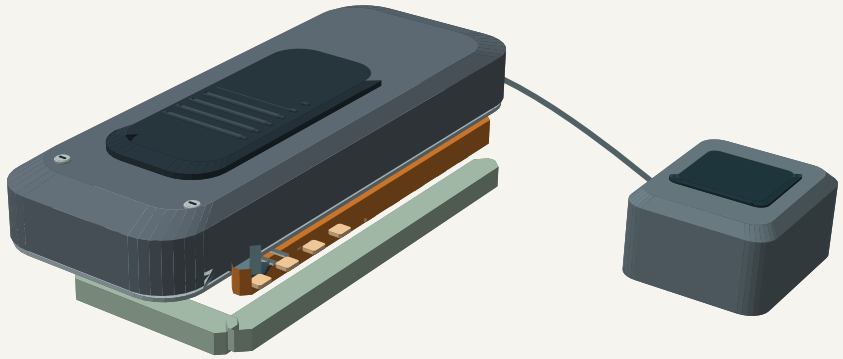
Clinician interprets anatomy and invalid or missing signal.

TARGET: synchronized acquisition, registration and explicit quality feedback. Added value, latency, exposure and thermal budgets remain UNVERIFIED.

S01 / SOURCE S02 / SOURCE S03 / SOURCE

A probe within a larger research system.

Illustrative geometry from the interactive VASCOUT model. No built prototype or measured packaging performance.



01 / HANDLE

Deliberate contact

Grip, controls and proposed force feedback need usability studies.

02 / INTERFACE

Passive contact shoe

Open corridor; acoustic and optical transmission require measurement.

03 / SYSTEM

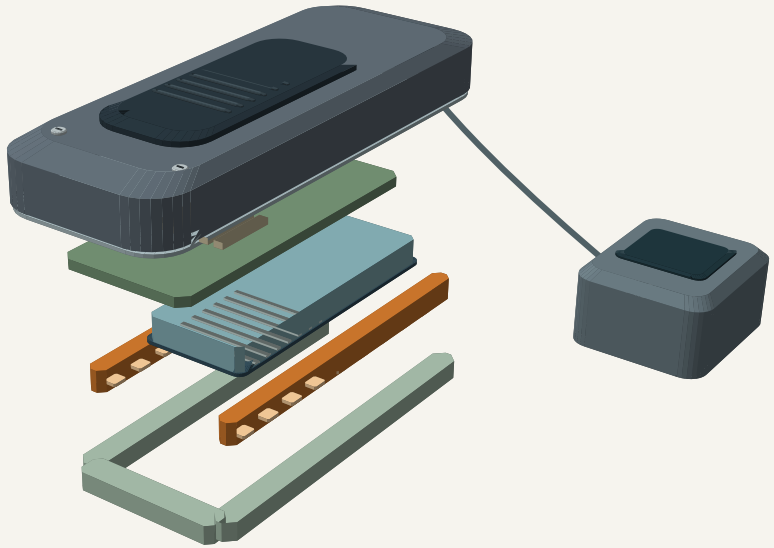
Separate acquisition unit

The source, acquisition chain and power budget are not contained in the handle.

TARGET / ILLUSTRATIVE / NOT TO SCALE

Five interfaces to engineer and test.

Layer separation is illustrative; materials, interfaces and integration remain design work.



01 Housing

Gloved reach, cleaning, balance and contact-force usability studies.

02 Acoustic sensor

Point-spread function, vessel localization, near-field performance and deformation.

03 Optical rails

Incremental benefit over US alone, exposure, heating, pigmentation effects and motion.

04 Contact shoe

Transmission loss, air gaps, vessel compression, cleaning and sterility requirements.

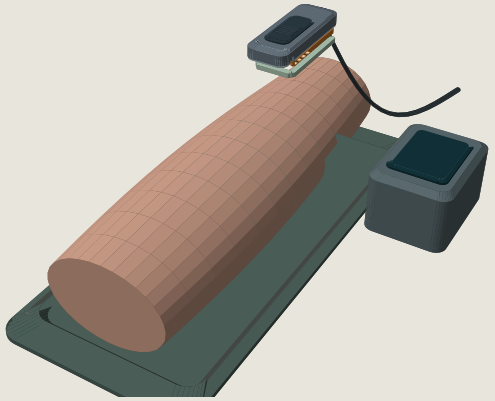
05 Electronics

Synchronization, registration, failure states, latency, power and thermal budgets.

SEPARATE SOURCE / ACQUISITION UNIT REQUIRED

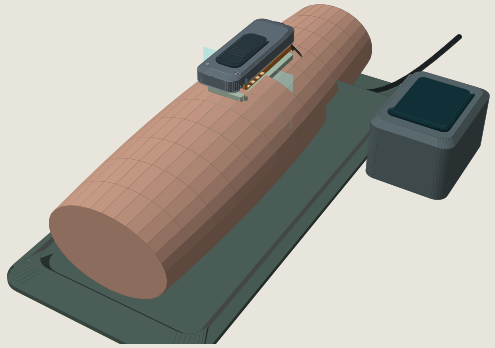
From a sweep to clinician review.

Static frames from a proposed 30-second workflow. Synthetic anatomy, pre-authored vessels and no measured patient data.



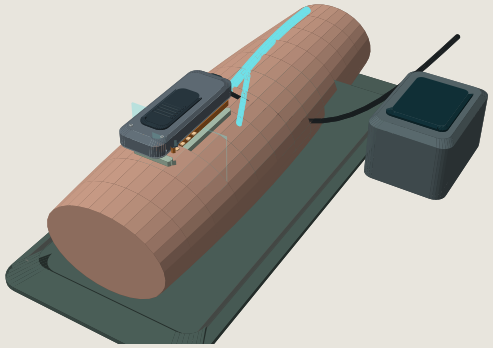
00s / FIELD SETUP

Proposed probe, tether and field pad.



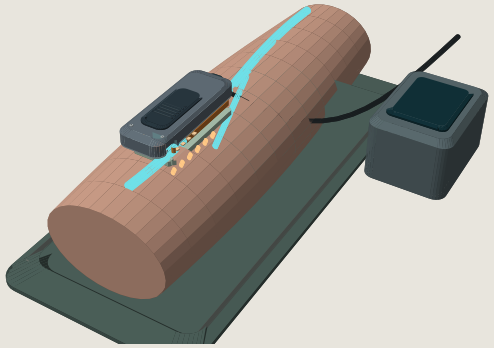
10s / SIMULATED SCAN

Drawn scan plane; no ultrasound acquisition.



19s / VESSEL MAP

Pre-authored path; no vessel detected.



23s / CLINICIAN REVIEW

Unresolved anatomy; no IV outcome.

CONCEPT ONLY: no scan generated, no vessel measured, no insertion guidance and no demonstrated access success.

Open the live animated concept

Numbers are inputs to investigation.

All values below are TARGETS or experiment inputs. They are not demonstrated device specifications.

Probe envelope	150 x 56 x 34 mm	Illustrative packaging; ergonomics and integration pending.
Ultrasound band	7-15 MHz candidate	Hardware selection and measured imaging response pending.
PA illumination	850 nm LED / 1064 nm pulsed	Alternative laboratory configurations; no combined claim.
Phantom depths	2 / 5 / 10 / 15 mm	Test positions; not validated operating depths.
Phantom diameters	1 / 2 / 4 / 6 mm	Test sizes; not clinical access thresholds.
3D acquisition	Tracked sweep / bench scan	Registration and motion sensitivity unmeasured.
Contact / invalid data	Passive shoe + force / UNKNOWN	Deformation, coupling, safety and failure modes to test.

NO DEPTH, FRAME-RATE, SAFETY OR FIELD-READY PERFORMANCE CLAIM

What VASCOUT must still establish.

Current position: concept design and literature review. Every device-level gate below is UNVERIFIED.

01 / IMAGING / UNVERIFIED

Resolution, localization and coverage

Benchmark against known geometry across the phantom matrix.

02 / HYBRID BENEFIT / UNVERIFIED

Incremental benefit over ultrasound

Define a task, comparator, success criteria and added time.

03 / EQUITY + QUALITY / UNVERIFIED

Pigmentation, motion and signal dropout

Measure failure modes; no skin-tone invariance established. [S06]

04 / CONTACT + SAFETY / UNVERIFIED

Deformation, exposure and heating

Measure optical/acoustic exposure, force and thermal budgets.

05 / FIELD USABILITY / UNVERIFIED

Gloves, cleaning and constrained use

Simulated use before any assertion of field suitability.

06 / CLINICAL + REGULATORY / UNVERIFIED

Intended use and evidence pathway

Expert review and appropriate approvals before human studies. [S11]

S06 / SOURCE

S11 / SOURCE

A path from concept to evidence.

Advancement depends on measured results and predefined failure criteria; no stage is marked complete.

01	02	03	04	05
BENCH COMPARISON	**CALIBRATED PHANTOMS**	**SIMULATED USE**	**EXPERT REVIEW**	**APPROVED HUMAN STUDIES**
Choose components, ground truth, calibration and US-only comparator.	Measure localization, diameter error, dropout, bias and deformation.	Test gloves, motion, lighting, cleaning, setup and failure recognition.	Refine intended use, risk controls and study requirements.	Only after appropriate approvals; evaluate safety and intended benefit.
UNVERIFIED	UNVERIFIED	UNVERIFIED	UNVERIFIED	UNVERIFIED

Decision at each gate: proceed, revise or stop based on agreed evidence. Foreign-body localization remains a separate future phantom study; negative imaging cannot exclude fragments. [S08, S09]

A first study to scope together.

LED-PA comparator and prior art / Proposed fit is an INFERENCE. No partner interest confirmed.

PROPOSED FIRST STUDY / TARGET

Compare a documented Acoustic X research configuration with ultrasound alone on matched phantoms; vary depth, diameter, optical attenuation, contact force and motion. Treat attenuation phantoms as limited models, not proof across human skin tones.

PROPOSED OUTPUT / TARGET

A comparator protocol and an incremental-benefit report that includes negative results, exposure/thermal measurements and acquisition overhead.

VALIDATION + ENGAGEMENT GAPS / UNVERIFIED

Equipment availability, configuration, data access, research support and willingness to collaborate are unverified. Related prior work means novelty cannot be assumed; no field-access outcome follows.

Explore the device and evidence library

DRAFT RESEARCH DISCUSSION / NOT SENT

CYBERDYNE / Acoustic X

LED-PA comparator and prior art / Proposed research fit is an INFERENCE. Partner interest is unconfirmed.

SOURCED CAPABILITY / P03, P04

CYBERDYNE presents Acoustic X as an LED-based PA/US research system. Its page already lists vascular-access-related work and provides an Acoustic X inquiry category.

PROPOSED FIRST STUDY / INFERENCE

Compare a documented Acoustic X research configuration with ultrasound alone on matched phantoms; vary depth, diameter, optical attenuation, contact force and motion. Treat attenuation phantoms as limited models, not proof across human skin tones.

PROPOSED OUTPUT

A comparator protocol and an incremental-benefit report that includes negative results, exposure/thermal measurements and acquisition overhead.

VALIDATION + ENGAGEMENT GAPS / UNVERIFIED

Equipment availability, configuration, data access, research support and willingness to collaborate are unverified. Related prior work means novelty cannot be assumed; no field-access outcome follows.

PUBLISHED ROUTE / NO OUTREACH SENT

Official product inquiry; select Acoustic X and request research-team routing.

P03 / OFFICIAL SOURCE

P04 / OFFICIAL SOURCE

TAILORED DRAFT PITCH / PREPARED WITH AI / NOT SENT

Your Acoustic X work already addresses LED-based PA/US and peripheral vascular targets, making it a relevant starting point for VASCOUT rather than evidence that the hybrid idea is new. Would your research team explore a comparator study focused on contact deformation, optical attenuation, motion and explicit invalid-signal reporting? We would agree a US-only baseline and measure incremental localization value, exposure, heating and added time. VASCOUT currently consists of a concept dossier and synthetic demonstration; the proposed collaboration would determine whether its interface and uncertainty-feedback ideas merit further development.

Official engagement route

CLINICAL BENEFIT / SAFETY / FIELD SUITABILITY: UNVERIFIED

Sources / clinical and imaging context

Research snapshot: 20 October 2026. URLs link to the retained source record; source evidence does not validate VASCOUT.

S01 / Current DFU / Manufacturer

Veinlite LEDX - directions for use

Manufacturer DFU. Visible transillumination; not NIR or PA tomography. No comparative efficacy proof.

S02 / 2024 / Healthy human

Photoacoustic vector tomography for deep haemodynamic imaging

Reported healthy-human vascular imaging. Different research system; full-text access incomplete.

S03 / 2021 / Healthy human

Photoacoustic imaging of human vasculature using LED versus laser illumination

LED/laser research comparison in phantoms and healthy limbs. No handheld field validation; publisher access limited.

S04 / 2022 / Clinical

Monitoring peripheral hemodynamic response to changes in blood pressure via photoacoustic imaging

Dialysis/exercise hemodynamics. Not a dehydration meter or shock classifier; root full-text reread incomplete.

S05 / 2022 / Phantom

Thin ceramic PZT dual- and multi-frequency pMUT arrays for photoacoustic imaging

Sensor-array prior art in agar. No real-time human access proof; direct PMC access challenged.

S06 / 2026 / Healthy human

The confounding effects of skin colour in photoacoustic imaging

42 healthy participants; optical/acoustic confounding. No demonstrated VASCOUT skin-tone invariance.

S10a / 2005 / Clinical

Ultrasonography-guided peripheral intravenous access versus traditional approaches

60 difficult-access patients; reported benefit. Odd/even-day allocation, two operators; full text unverified. Read with S10b.

S10b / 2009 / Clinical

Ultrasonographically guided peripheral intravenous cannulation: a randomized trial

59-patient randomized ED trial; no demonstrated superiority. Workflow-specific; full text unverified. Read with S10a.

Sources / limitations, engineering and prior art

Research snapshot: 20 October 2026. URLs link to the retained source record; source evidence does not validate VASCOUT.

S07 / 2026 / Guideline

TCCC Guidelines - vascular access claim audit

1 May 2026 guideline, directly read in the retained audit. Clinical context only; no treatment directions in this deck.

S08 / 2009 / Cadaver

Bedside ultrasound for detection of soft tissue foreign bodies: a cadaveric study

Cadaver study; small objects and poor reliability. Negative imaging cannot universally exclude fragments.

S09 / 2018 / Simulation

Ultrasound evaluation of soft-tissue foreign bodies by US Army medics

28 medics, chicken-thigh simulation after training. No injured-human or battlefield validation; retrieval limits retained.

S11 / 2021 / Regulator

FDA: Imagio Breast Imaging System - P200003

FDA approval for another product and breast-imaging indication. No VASCOUT clearance or pathway follows.

S12 / 2018 / Healthy human

A device for improving the visual clarity and dimension of veins

20 healthy volunteers in vacuum-device research. Not shock evidence; retrieval gaps. Suction excluded here.

L01 / Manufacturer / Manufacturer

TI AFE5832LP - ultrasound analog front end

Manufacturer: 32 receive inputs, 16 multiplexed ADC channels. Complete system architecture unresolved.

L02 / Manufacturer / Manufacturer

Analog Devices MAX2082 - octal ultrasound transceiver

Manufacturer: octal transmit/receive device. Does not prove proposed system integration or power budget.

G01 / 2026 / Concept material

User-supplied Gemini concepts - audited research inputs

User-supplied Gemini proposals, not studies. Audited concept inputs; no scientific validation. See live evidence library.

Sources / this research prospect

Accessed 20 October 2026 EDT / 21 October UTC. Organization capability and route evidence; not VASCOUT validation.

P03

CYBERDYNE - Acoustic X research system

Direct official English page read, reached through its language link

Manufacturer describes an LED-based PA/US research system and lists a webinar and prior publication about peripheral vascular targets.

LIMIT: Product descriptions and linked-study summaries are not an independent efficacy audit; no VASCOUT novelty or field validation established.

P04

CYBERDYNE - Contact

Direct official page read

The official inquiry form has a product category for Acoustic X.

LIMIT: Product inquiry route, not an open research award or confirmed collaboration.

PUBLIC SOURCES / PROPOSED FIT IS INFERENCE / INTEREST UNCONFIRMED